

Read and identify key facts & relationships (step 1). 10 min.
阅读理解。提示：寻找指代词、重复关键词、关联词等线索。

We are in the midst of a once-in-a-generation opportunity to remake our approach to science. This moment, in all its difficult uncertainty—COVID-19, economic turmoil among others—demonstrates why universities and other research stakeholders should move decisively to embrace open science. By adopting what are called “open science” practices, we can align the incentive structures of research production and consumption with our values, and catalyze the scientific progress our society so desperately needs. Open science, according to Michael Nielson, is “the idea that scientific knowledge of all kinds should be openly shared as early as is practical in the discovery process.” That open science is an integral tool in the fight against COVID-19 is indisputable: the importance of access to scientific articles and data to help identify promising vaccines and therapeutics was recognized by publishers and researchers alike early in the pandemic. As a consequence, the research community has worked rapidly to take down the barriers that chronically impede scientific progress.

What is critical thinking? ✓

Critical thinking is **all about asking questions**, the right questions, questions that help you **assess both the meaning and the significance of claims and arguments**. Building these skills and applying them in your life makes it easier for you to assess evidence, evaluate arguments and adapt your thinking, so you **stay switched on and engaged in different situations**. Critical thinking **involves stepping back from a situation to enable you to see all the angles before making judgments or taking decisions**. It means identifying the key points, analyzing the sources of information, weighing up different types of evidence, just as a judge and jury would do in a court of law and putting it all together into your own independent thought through point of view. One thing that it's very important to realize is a critical thinking **isn't about being critical and it's about much more than just finding flaws in other people's claims**. By itself that isn't enough to **give you an edge**. To be a true critical thinker means being **creative, reflective, and adaptable**, evaluating the evidence to decide for yourself what is accurate, what is relevant and do I have sufficient information to take a decision on this topic. Thinking critically means **taking a stand for yourself**. It can be difficult not to be swayed by close family or friends views on things, or certain beliefs that just feel right, but learning how to **use these higher-order thinking skills** can help you to feel much more confident in your own opinions and conclusions. ✓

Critical thinking is **also about a sense of discovery and excitement**, not only about learning but evaluating arguments to see how they stand up and **filtering for yourself what resonates is right or wrong**. By using these techniques you'll find yourself becoming a clearer better thinker. | ✓



Survey the text and fill in the outline

I. Horgan's purpose of teaching critical thinking (para.s 1-3):

II. Teaching content_part 1 (para.s 4-7): _____

III. Teaching content_part 2 (para.s 8-12): _____

IV. Teaching outcome (para.s 13-15): _____

V. Horgan's response to students (para. 16)

VI. Ending (para. 17): corresponding the title ("backfire")





Read the text again and fill in the blanks using words from the text.

John Horgan has been teaching a critical thinking course for all freshmen at Stevens Institute of Technology. His original plan is to give students uncertainty, doubt and skepticism and raise their awareness of science' s fallibility. His own awareness, according to Horgan, is acquired through effort. He used to be easily impressed by scientists' claims but gradually his faith got undermined. The prospects for future depicted by scientists and journalists are overly optimistic. In fact, science is not as potent or fast-moving.



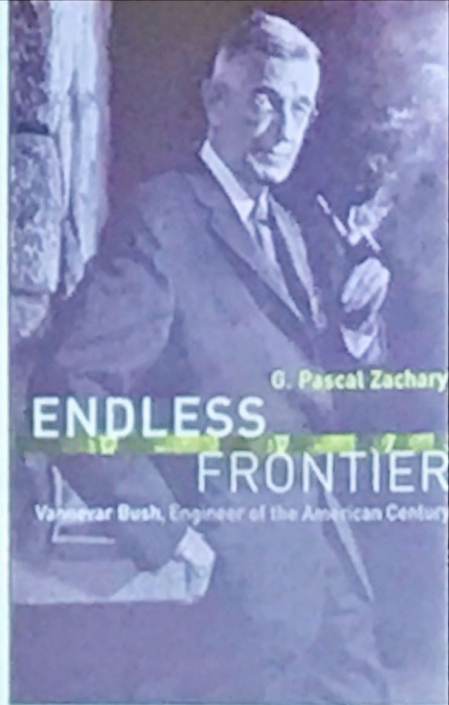


Read the text again and fill in the blanks using words from the text.

John Horgan has been teaching a critical thinking course for all freshmen at Stevens Institute of Technology. His original plan is to give students uncertainty, doubt and skepticism and raise their awareness of science's fallibility. His own awareness, according to Horgan, is acquired through effort. He used to be easily impressed by scientists' claims but gradually his faith got undermined. The prospects for future depicted by scientists and journalists are overly optimistic. In fact, science is not as potent or fast-moving.



As a young professor at MIT in the 1920s, Vannevar Bush (1890-1974) did seminal work on analog computing and was a cofounder of Raytheon, whose initial success was based on long-lasting radio tubes. But he is best known for his role in Washington during World War II: as President Roosevelt's advisor, he organized the Manhattan Project and oversaw the work of 6,000 civilian scientists designing new weapons. His 1945 report "Science -- The Endless Frontier" spurred the creation of a system of public support for university research that endures to this day. Although he helped to give rise to the military-industrial complex, Bush was a skeptical observer of the interplay between science and politics. He warned against the dangers of an arms race and led a failed effort to halt testing of the hydrogen bomb. This balanced and gracefully written biography brings to life an American original and his times.

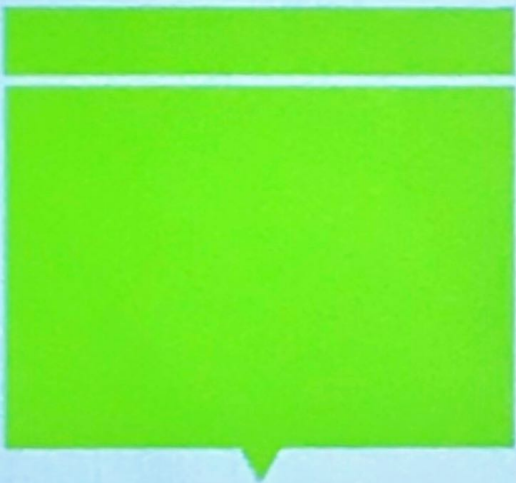




Read the text and fill in the blanks **without** using expressions from the text. Paraphrasing is required.

To prove his point, Horgan analyzes claims from fields such as physics, cosmology, neuroscience, genetics and artificial intelligence. He also asks students to read other skeptics, including Gary Taubes, John Ioannidis and Philip Tetlock. Both Taubes and Ioannidis suggested that the latest, most dramatic research findings in epidemiology should not be trusted, due to the implausibility of most peer-reviewed research. Ioannidis found a correlation between the fame of experts and their unreliability. So did Tetlock. Based on his 20-year study, Tetlock concluded that expertise should not be trusted. Intelligent as experts are, they are good at data mining* and may as well be deluded.





中国人有穷变通久的哲学，阿拉伯人也说“没有不变的常态”。我们尊重阿拉伯国家的变革诉求，支持阿拉伯国家自主探索发展道路。处理好改革发展稳定关系十分重要。这就好比阿拉伯喜闻乐见的赛骆驼，前半程跑得太快，后半程就可能体力透支；前半程跑得太慢，后半程有可能跟不上。骑手只有平衡好速度和耐力，才能够坚持到最后。

推动中阿两大民族复兴形成更多交汇

Translate

坚持自我，最大的障碍之一就是太在意别人的眼光。要坚持自我，你必须甘愿承受有负众望、惹人非议的风险。倘若太过执着于别人的眼光，不愿意承担如此风险，那么只能沦为傀儡，无法真正做到坚持自我。一个人可以做的最有用的事情就是做自己，而做自己就意味着克服自己的问题，只有真正克服自己的问题，才能帮助别人克服他们的问题。如果你能够做出有价值的贡献，别人的看法根本无足轻重。

But in this age of climate crisis debates are renewed.



Negative side (con): mainly business leaders & politicians

Their evidence:

- ✧ Growth is an economic necessity. ➡ Degrowth is not only anti-consumerist (i.e. undermining business) but with dire social consequences (i.e. affecting social functioning)
- ✧ What's needed is for growth to be more eco-friendly.
- ✧ Current growth patterns can be maintained through product innovations (i.e. making products more eco-friendly)

Affirmative side (pro): many major figures, e.g. Noam Chomsky, Yanis Varoufakis, Anthony Giddens.

The basic structure of an argumentation

Introduction

Hook+issue
Thesis Statement

Topic sentence 1 (sub topic 1)
Specific evidence +analysis

Topic sentence 2 (sub topic 2)
Specific evidence +analysis

Topic sentence 3 (refutation)
Specific evidence +analysis

Body

Conclusion

Restate Thesis Statement
Closing Remarks/Appeal

How to write a good thesis?

➤ 2. Be assertive and develop one stance

Sample 1: This paper discusses paper bags and plastic ones.

Sample 2: Paper bags are better than plastic ones for use in ... because

How to write a good thesis?

- **4. Clear and relevant sub-points to show why your stance is worth noting**

Sample :

Paper bags are better than plastic ones for use in grocery stores because making them requires a smaller carbon footprint, they are cheaper to buy, and they are significantly easier to recycle.

Gold, a precious metal, is prized for two important characteristics. First of all, gold has a lustrous beauty that is resistant to corrosion. Therefore, it is suitable for jewelry, coins and ornamental purposes. Gold never needs to be polished and will remain beautiful forever. For example, a Macedonian coin remains as untarnished today as the day it was minted twenty-three centuries ago. Another important characteristic of gold is its usefulness to industry and science. For many years, it has been used in hundreds of industrial applications. The most recent use of gold is in astronauts' suits. Astronauts wear gold-plated heat shields for protection outside the spaceship. In conclusion, gold is treasured not only for its beauty, but also for its utility.

It has been argued that dolphin parks provide the only opportunity for much of the public to see marine mammals (Smith, 1992). Most Australians, *so this argument goes*, live in cities and never get to see these animals. *It is claimed that* marine parks allow the average Australian to appreciate our marine wildlife. *However, as* Smith *states*, dolphins, whales and seals can be viewed in the wild at a number of places on the Australian coast. *In fact*, there are more places where they can be seen in the wild than places where they can be seen in captivity. *Moreover*, most Australians would have to travel less to get to these locations than they would to get to the marine parks on the Gold Coast. *In addition*, places where there are wild marine mammals do not charge an exorbitant entry fee - they are free.

Dr Alison Lane, the director of the Cairns Marine Science Institute, *contends that* we need marine parks for scientific research (The Age, 19.2.93). She *argues that* much of our knowledge of marine mammals comes from studies which were undertaken at marine parks. The knowledge which is obtained at marine parks, *so this argument goes*, can be useful for planning for the conservation of marine mammal species. *However, as* Jones (1991) *explains*, park research is only useful for understanding captive animals and is not useful for learning about animals in the wild. Dolphin and whale biology changes in marine park conditions. Their diets are different, they have significantly lower life spans and they are more prone to disease. *In addition*, marine mammals in dolphin parks are trained and this means that their patterns of social behaviour are changed. Therefore research undertaken at marine parks is generally not reliable.

It is the contention of the Marine Park Owners Association *that* marine parks attract a lot of foreign tourists (The Sun-Herald 12.4.93). *This position goes on to assert that* these tourists spend a lot of money, increasing our foreign exchange earnings and assisting our national balance of payments. *However*, foreign tourists would still come to Australia if the parks were closed down. *Indeed*, surveys of overseas tourists show that they come here for a variety of other reasons and not to visit places like Seaworld (The Age, Good Weekend 16.8.93). Tourists come here to see our native wildlife in its natural environment and not to see it in cages and cement pools. They can see animals in those condition in their own countries *Furthermore*, we should be promoting our beautiful natural environment to tourists and not the ugly concrete marine park venues.

Translation Exercise

现在有一种说法是人文学科已经过时。当前就业市场形势如此严峻，很多大学生及学生家长都想要选择那些将来容易就业的专业，而对人文学科嗤之以鼻。这种想法显然过于狭隘。人文学科着眼的是培养通才。它不像自然科学和社会科学那样重视普遍规律和知识的积累，而是更注重一种智慧和能力的培养。它更注重有创造性的探讨过程而非固定的结论。它对个人的独创性留有更大的空间。

A Suggested Translation Version

- It is said that the humanities are out of date. Given the tough job market nowadays, they are no longer appealing in the eyes of many college students and their parents who want to pick those majors that will help land a job in the future. This idea is obviously too narrow-minded. The purpose of the humanities is to make students well-rounded. Compared to the natural and social sciences, the humanities value more the cultivation of wisdom and competence, rather than the accumulation of general rules and knowledge. They appreciate more innovative deliberation rather than some fixed conclusions. As such, they leave more room for originality.